Book II of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Document Status

This volume is the technical documentation on formal reconstruction of Book II of Euclid's *Elements*. It records the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable rectangle and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

The documentation accompanies the formal development. The Lean source is authoritative for theorem statements, hypotheses, dependencies, and proof status.

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1.1 Introduction

Proposition II.1 opens Book II with a change of mathematical language. Book I has developed a synthetic theory of congruence, parallels, parallelograms, squares, and equal content. Book II begins to combine these ingredients in a systematic calculus of rectangles. A modern reader immediately recognizes in II.1 the pattern of a distributive law. That recognition is useful, but it must be stated carefully: Euclid does not introduce variables for numerical lengths, does not define multiplication of lengths, and does not prove an identity in a number field.

The primitive object in the proposition is a geometric figure: a rectangle "contained by" two straight lines. When one of the two lines is divided into parts, the large rectangle is divided into smaller rectangles having the same other side. The equality asserted by the proposition is therefore first of all a statement about decomposition of figures.

This is a natural point at which to speak of the beginning of the *geometrical algebra* characteristic of Book II. The phrase is an interpretive description of a stable pattern of geometric operations, not a claim that Euclid was secretly manipulating modern symbolic formulas. In the present formal reconstruction this distinction is especially sharp: the code contains rectangles, segment congruence, betweenness, parallelograms, right angles, and scissors relations, but no multiplication operation on segment lengths and no numerical area function.

The formal development separates three levels:

1. a concrete rectangle and its geometric cuts;

2. the Euclidean phrase "rectangle contained by" two given segments;

3. independence of the particular rectangle chosen to represent that phrase.

The final theorem is stronger than a bare equal-content statement. Its conclusion is the exact relation

```
HilbertScissorsEq
```

for the whole rectangle and the formal sum of its parts.

1.2 Euclid's "Rectangle Contained By"

1.2.1 The original geometric meaning

Book II begins with the terminology that a rectangular parallelogram is "contained by" the two straight lines which contain one of its right angles. Thus the expression does not initially denote a product of two numbers. It names a rectangle by two adjacent sides.

Suppose a rectangle has adjacent sides congruent to two given segments PQ and RS. Euclid's phrase

the rectangle contained by PQ, RS

should be read geometrically: choose a right-angled parallelogram whose two adjacent sides have those magnitudes. The rectangle itself remains a plane figure.

For exposition only, we will occasionally write

$$\mathrm{Rect}(PQ, RS).$$

This is documentary shorthand. It is not a Lean definition, it is not a number-valued function, and the comma is not multiplication.

1.2.2 Why this already looks algebraic

If BC is divided into consecutive parts BM, MN, and NC, II.1 says in this shorthand

$$\mathrm{Rect}(BC, PQ) \;=\; \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MN, PQ) + \mathrm{Rect}(NC, PQ).$$

A modern algebraist naturally sees the shape

$$(b_1 + b_2 + b_3)a = b_1 a + b_2 a + b_3 a.$$

The formal proof, however, establishes the geometric statement on the first line, not the numerical identity on the second. The second formula is only a modern mnemonic for the combinatorial pattern of the dissection.

This is consistent with the modern synthetic reading of Euclid's area theory. Hartshorne treats Book II as a collection of equal-content statements about figures and describes the resulting manipulation as "geometrical algebra." The point is precisely that the manipulation precedes any assignment of numerical areas to the figures.

1.3 The Synthetic Rectangle Layer

1.3.1 Rectangles

The reconstruction introduces a small reusable rectangle API in `HilbertRectangle.lean`. A rectangle is a parallelogram with one specified right angle:

```
def IsRectangle
    (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  HilbertRightAngle Geo A B C
```

The parallelogram structure supplies the opposite-side relations, while the right-angle component distinguishes rectangles from arbitrary parallelograms. The remaining right angles can then be propagated synthetically.

1.3.2 The formal meaning of "contained by"

The Euclidean phrase is represented by the predicate

```
def IsRectangleContainedBy
    (A B C D P Q R S : Geo.Point) : Prop :=
  IsRectangle Geo A B C D /\
  Geo.Congruent A B P Q /\
  Geo.Congruent B C R S
```

Thus

```
IsRectangleContainedBy Geo A B C D P Q R S
```

means exactly that $ABCD$ is a concrete rectangle, $AB \cong PQ$, and $BC \cong RS$. Nothing in this definition assigns a number to PQ or RS. The two given segments specify the two adjacent geometric magnitudes of a representative rectangle.

The layer also proves existence of such a representative and, more importantly for the semantics of Book II, uniqueness of its scissors content. The theorem

```
rectangle_contained_by_unique
```

says that any two rectangles contained by the same ordered pair of segments satisfy the exact scissors relation. Consequently Euclid's definite description "the rectangle contained by PQ, RS" becomes well-defined at the scissors level even though many different concrete quadrilaterals can represent it.

1.3.3 No segment multiplication and no numerical area

There are two tempting modernizations that the present reconstruction explicitly avoids.

First, we do not define a product of segment lengths $|PQ|\,|RS|$. The pair (PQ, RS) remains a pair of geometric segments, and the rectangle is related to that pair by congruence of adjacent sides.

Second, we do not map a rectangle to a real number called its area. The content of the proposition is expressed through relations between finite formal sums of triangles. This permits addition, dissection, transport, and comparison of figures at the synthetic level.

This is also conceptually distinct from Hilbert's later arithmetic of segments, where arithmetic operations are introduced as an additional construction. No such field or segment-arithmetic layer is imported into II.1.

1.4 Exact Scissors Equality and Equal Content

The project uses two related but different content relations. They should not be conflated.

1.4.1 Exact scissors equality

`HilbertScissorsEq` records an exact finite cut-and-paste relation between formal scissors terms. In II.1 it is produced by explicit splits of the rectangle into smaller rectangles and by normalization of the triangulated parallelogram terms. Schematically,

$$P \simeq_{\mathrm{sc}} Q + R$$

means that the two sides can be matched by the exact scissors machinery already developed in the project.

The rectangle transport theorem is already available in this stronger form:

```
rectangle_transport_scissorsEq
```

for rectangles with congruent corresponding adjacent sides.

1.4.2 The more general equal-content relation

The broader project notion corresponding to equal content is

```
HilbertScissorsEquicomplementable
```

It allows suitable complementary terms to be adjoined before exact scissors equality is established. Thus it is designed to capture a more general synthetic equal-content relation.

The theorem

```
rectangle_transport
```

is obtained by weakening the exact transport theorem from `HilbertScissorsEq` to equicomplementability.

For II.1 no such weakening is necessary. The final theorem proves `HilbertScissorsEq` itself. Therefore II.1 certainly yields an equal-content consequence, but its formal statement retains the stronger fact that the displayed equality comes from an exact dissection.

1.5 The Classical and Formal Configurations

1.5.1 The binary split

The reusable core is a one-cut theorem. Let $B - M - C$ and let the corresponding point L divide the opposite side DE. The rectangle $BCED$ is cut into the rectangles $BMLD$ and $MCEL$:

$$BCED \simeq_{\mathrm{sc}} BMLD + MCEL.$$

In Lean the theorem is

```
rectangle_split
```

and it returns both the rectangle structure of the two pieces and the exact scissors equality.

The formal configuration contains one additional point X. It satisfies $D - X - C$ and $M - X - L$. This is not decorative data: the inherited exact split lemma from the I.47 scissors infrastructure uses the intersection of the cut line with a diagonal. The classical rectangle picture suppresses this witness; the formal theorem makes it explicit.

1.5.2 The three-part split

The three-part form applies the binary split twice. First cut at $M - L$; then cut the remaining rectangle at $N - K$. The result is

$$\begin{aligned} BCED \simeq_{\text{sc}} BMLD \\ + MNKL \\ + NCEK. \end{aligned}$$

The second split has its own diagonal-intersection witness Y, with $L - Y - C$ and $N - Y - K$.

Figure 1.1 shows both levels and includes the hidden witnesses X and Y required by the current Lean interface.

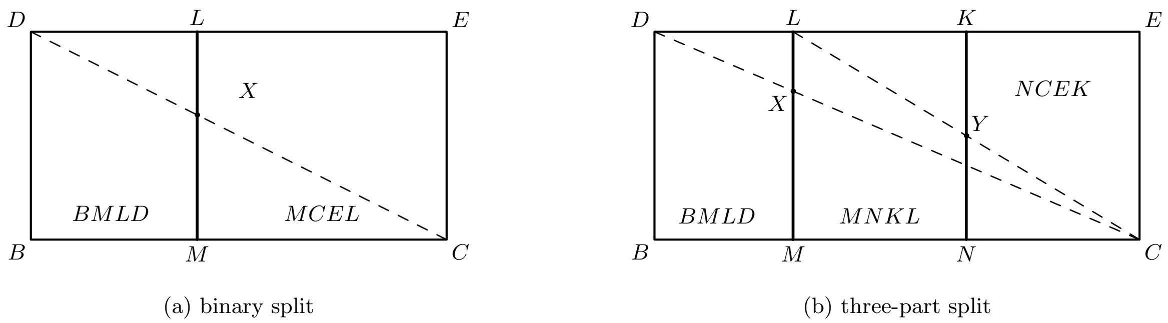


Figure 1.1: The exact rectangle splits underlying II.1. Panel (a) is the binary kernel: $B - M - C$ and the cut ML divide $BCED$ into $BMLD$ and $MCEL$; the dashed diagonal DC meets the cut at the formal witness X. Panel (b) applies the same construction twice: BC is divided into BM, MN, and NC, producing the three rectangles $BMLD$, $MNKL$, and $NCEK$. The dashed diagonals display the witnesses X and Y used by the two exact split calls.

The actual three-part theorem is

```
rectangle_split_three
```

Its proof is structurally transparent: obtain the first binary equality, split the remaining right-hand rectangle, add the unchanged left piece to that second equality, and compose by transitivity of `HilbertScissorsEq`.

1.5.3 Scope of the present formal statement

Euclid states II.1 for one line cut into an arbitrary finite number of parts. The current Lean file does not yet package an n-ary list-valued theorem. It proves the binary split and the explicit three-part form used in the standard II.1 configuration. The binary theorem is the reusable kernel from which a general finite version could later be obtained by iteration.

This is a deliberate statement-fidelity point. The documentation records Euclid's general formulation, but the final formal statement displayed below is the representative-independent three-part theorem actually present in `Proposition2_1.lean`.

1.6 From a Concrete Diagram to Euclid's Language

The geometric split alone produces concrete rectangles. Euclid II.1 is more abstract: it speaks of the rectangles *contained by* the whole segment and the several parts together with one fixed segment.

The theorem

```
rectangle_split_three_contained_by
```

bridges these levels. If $CE \cong PQ$, it packages the four concrete rectangles as

$$\begin{array}{rcl} BCED & : & \mathrm{Rect}(BC, PQ), \\ BMLD & : & \mathrm{Rect}(BM, PQ), \\ MNKL & : & \mathrm{Rect}(MN, PQ), \\ NCEK & : & \mathrm{Rect}(NC, PQ). \end{array}$$

The opposite-side congruences of the parallelograms are what transport the fixed side PQ through the two cuts. The conclusion still contains the same exact scissors equality between the concrete quadrilateral terms.

At this stage the geometry corresponding to Euclid's wording is visible, but the theorem still uses one particular drawn rectangle and its particular cut points. The final step removes this representational dependence.

1.7 Representative Independence

1.7.1 Why representatives matter formally

A phrase such as

$$\mathrm{Rect}(BC, PQ)$$

does not identify four vertices. Lean, however, must manipulate an actual term such as

```
hilbertParallelogramTerm Geo U0 U1 U2 U3
```

with named vertices. The final theorem therefore accepts arbitrary representatives for the whole rectangle and for each of the three part rectangles.

The concrete dissection first proves

$$BCED \simeq_{\mathrm{sc}} BMLD + MNKL + NCEK.$$

Then four calls to

```
rectangle_contained_by_unique
```

transport the whole and the three pieces to arbitrary representatives U, V, W, and Z. Compatibility of `HilbertScissorsEq` with formal addition transports the nested sum. Transitivity completes the chain

$$\begin{gathered}\text{arbitrary whole representative}\\ \Downarrow\\ \text{concrete whole rectangle}\\ \Downarrow\\ \text{concrete three-part dissection}\\ \Downarrow\\ \text{arbitrary representatives of the three parts.}\end{gathered}$$

This is the formal content of representative independence. It is not merely a convenience for Lean. It explains why the Euclidean expression "the rectangle contained by two segments" can function as a stable geometric magnitude without first quotienting segments into numerical lengths or assigning numerical areas to rectangles.

1.8 Commutativity by Rectangle Rotation

The second algebra-like feature is symmetry in the two containing segments. For a numerical product one would write $ab = ba$. The synthetic proof does not appeal to a commutative multiplication law. Instead it rotates the presentation of a rectangle.

If $ABCD$ is contained by PQ, RS, then the cyclically reoriented rectangle $BCDA$ has adjacent sides BC and CD. The first is already congruent to RS; the second is congruent to AB, hence to PQ. Thus the same geometric rectangle, with its vertices read from the next corner, is contained by RS, PQ.

The exact theorem available in the rectangle layer is

```
rectangle_contained_by_swap
```

It returns both the swapped `IsRectangleContainedBy` fact and a `HilbertScissorsEq` identifying the two quadrilateral presentations. The current rectangle API also packages the representative-independent consequence as

```
rectangle_contained_by_comm
```

so the commutativity needed by later Book II propositions is a theorem of rectangle geometry rather than an assumed law of segment multiplication.

Figure 1.2 shows the geometric mechanism. The operation is a cyclic change of the distinguished corner of the same rectangle; this exchanges the two adjacent side roles.

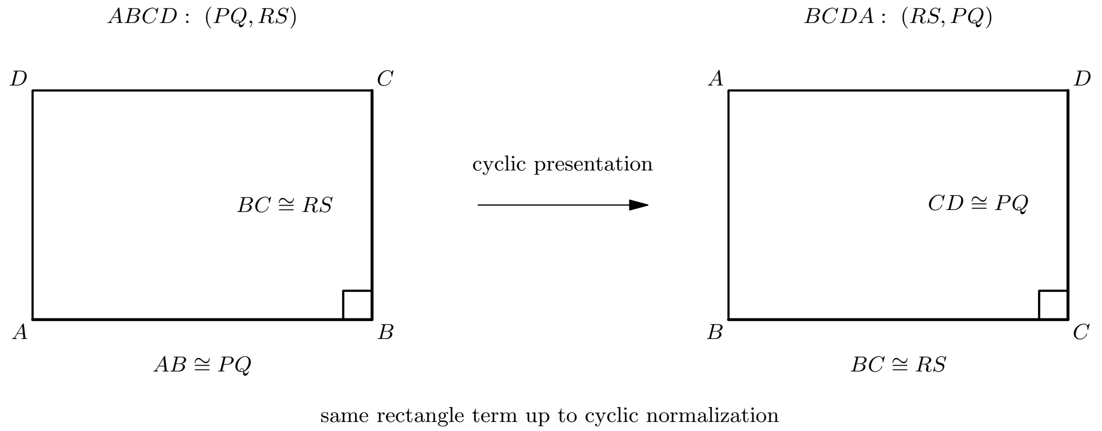


Figure 1.2: The geometric source of commutativity. On the left, $ABCD$ represents the rectangle contained by PQ, RS. Reading the same rectangle from the next vertex gives $BCDA$: now the first adjacent side represents RS and the second represents PQ. Exact scissors invariance under this cyclic presentation change yields the synthetic counterpart of exchanging the two factors.

This lemma is already operational in the next proposition. II.2 naturally writes rectangles in the order "whole side, part", whereas the II.1 cut lemma produces them in the order "part, fixed side". The implementation of II.2 uses the swap theorem to pass between these two presentations before applying the binary form of II.1.

1.9 Proof Architecture of the Reconstruction

The Lean proof is best read as five mathematical stages.

1.9.1 Step 1: exact binary dissection

```
rectangle_split_scissorsEq
rectangle_split_left
rectangle_split_right
rectangle_split
```

The inherited I.47 split theorem supplies the exact scissors decomposition. The two local rectangle helpers prove that each piece remains a rectangle by propagating right angles through the relevant parallelograms and along the same rays determined by betweenness.

1.9.2 Step 2: iterate the split

```
rectangle_split_three
```

The right-hand remainder of the first split is itself a rectangle, so the same binary theorem can be applied again. Addition and transitivity of exact scissors equality compose the two decompositions.

1.9.3 Step 3: attach the "contained by" data

```
rectangle_split_three_contained_by
```

Opposite-side congruence transports the fixed segment PQ through all three pieces. Each concrete piece is thereby promoted from `IsRectangle` to `IsRectangleContainedBy`.

1.9.4 Step 4: remove the choice of representatives

```
rectangle_contained_by_unique
euclid_proposition_2_1_three
```

The uniqueness theorem replaces the concrete whole and each concrete part by an arbitrary rectangle contained by the same pair of segments. This is the semantic step that makes the final statement faithful to Euclid's definite phrase "the rectangle contained by."

1.9.5 Step 5: expose the symmetric side order

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

Cyclic rotation of the rectangle exchanges the two containing segments. This is not needed to establish the displayed three-part split itself, but it is an important part of the Book II rectangle interface and is immediately useful in II.2 and later propositions.

1.10 Classical Proof and Formal Proof

1.10.1 Euclid's proof pattern

In the classical three-part configuration, take one undivided segment A and a segment BC divided at D and E. Erect a perpendicular at one endpoint of BC, lay off on it a segment equal to A, complete the outer rectangle, and through the division points draw parallels to the perpendicular side. The outer rectangle is visibly partitioned into three rectangles. Each smaller rectangle is contained by A and one of the three consecutive parts of BC. Common Notion 2 then combines the parts into the whole equality.

The classical DAG is therefore extremely small:

$$\begin{gathered}\text{divide one side}\\ \Downarrow\\ \text{draw the corresponding parallels}\\ \Downarrow\\ \text{obtain three subrectangles}\\ \Downarrow\\ \text{whole rectangle} = \text{sum of the three subrectangles.}\end{gathered}$$

1.10.2 The formal DAG

The current reconstruction expands two things that the classical diagram treats as immediate: exact scissors bookkeeping and independence of the chosen rectangle representatives.

$$\begin{gathered}\text{rectangle geometry + betweenness + cut parallelograms}\\ \Downarrow\\ \texttt{rectangle\_split}\\ \Downarrow\\ \texttt{rectangle\_split\_three}\\ \Downarrow\\ \texttt{rectangle\_split\_three\_contained\_by}\\ \Downarrow\\ \texttt{rectangle\_contained\_by\_unique}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_1\_three}.\end{gathered}$$

The proof is therefore not a mechanical transcription of Euclid. It preserves the geometric core, but makes the semantic and representational layers explicit.

1.11 Geometry, Content, and Representation

1.11.1 Geometry

The geometric data are substantial even though the proposition is visually simple. The current theorem receives the outer rectangle, strict betweenness relations on the divided side and its opposite, two diagonal-intersection witnesses, and four parallelogram facts for the two successive cuts. Right angles of the two resulting rectangles are not read from the picture; they are proved by right-angle transport and parallelogram geometry.

No coordinates are introduced. The cut points are controlled by `Between`, the side lengths by segment congruence, and the rectangular structure by parallels and synthetic right angles.

1.11.2 Content

Content enters in its strongest local form. The core statements are exact `HilbertScissorsEq` decompositions. The three-part theorem is composed from two exact binary splits by addition and transitivity. Representative transport is also exact scissors equality.

The more general equicomplementability relation is part of the surrounding Book I content theory, but II.1 does not need to weaken its conclusion to that level.

1.11.3 Representation

Representation is central to the formal meaning of the proposition. A rectangle contained by PQ, RS is not one canonical four-tuple of vertices. The predicate `IsRectangleContainedBy` describes a representative; `rectangle_contained_by_unique` proves that different representatives have the same exact scissors content.

There is a second representation issue inside the split itself. The inherited scissors theorem and the Book II statement use different cyclic orders for some parallelogram terms. The proof normalizes these by

```
parallelogram_term_rotateOne
```

before composing the scissors equalities. These cyclic changes are bookkeeping of a geometric representation, not new area facts.

1.12 Hidden Configuration Data

The classical drawing makes several facts appear automatic. Lean records them explicitly:

- $B - M - C$ for the first cut and $M - N - C$ for the second;
- $D - L - E$ and $L - K - E$ on the opposite side;
- $D - X - C$ together with $M - X - L$ for the first diagonal/cut intersection;
- $L - Y - C$ together with $N - Y - K$ for the second diagonal/cut intersection;
- explicit parallelogram witnesses for the left and right pieces of each split;
- congruence $CE \cong PQ$, which identifies the fixed side used in the "contained by" descriptions;
- four arbitrary rectangle representatives in the final theorem.

The points X and Y do not appear in Euclid's statement and need not be emphasized in an elementary picture of II.1. They are nevertheless real data of the current exact-dissection API, inherited from the triangulated proof of the rectangle split.

1.13 Axiomatic Status

The public II.1 theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so II.1 is Euclidean in the current project API. Group IV is available through this typeclass and is consumed transitively by the parallelogram and parallel infrastructure on which the rectangle layer and inherited split theorem depend. The proposition itself does not introduce a new parallel axiom.

No continuity typeclass, Archimedean axiom, numerical area function, or segment field is introduced in `Proposition2_1.lean`. No proposition-local `axiom` or `sorry` occurs in the audited source. The modular builds for `HilbertRectangle` and `Proposition2_1` are recorded as clean at the present project checkpoint.

The content status is also worth separating from the later converse arguments of Book I. II.1 proceeds from geometry to an exact dissection. It does not infer metric geometry from an equal-content hypothesis, so it does not need a proper-part or De Zolt-type strictness argument.

1.14 Relationship with Book I and the Following Propositions

The exact split theorem reuses the scissors infrastructure established during I.47. In particular, the underlying theorem historically named

```
i47_square_split
```

actually requires only parallelogram structure, and II.1 repackages it as a general rectangle split.

The rectangle layer also uses Book I results about parallelograms, opposite sides, congruent right angles, SAS/SSS transport, and the scissors representation of parallelograms. Thus Book II does not start a new foundational geometry; it starts a new way of organizing geometric magnitudes on top of the completed Book I infrastructure.

II.2 immediately imports II.1. Its formal proof illustrates why the symmetric rectangle API matters: Euclid's order of the two containing sides is opposite to the order produced by the binary cut theorem, so the representatives are swapped by rectangle rotation before II.1 is applied. II.3 continues the same rectangle calculus. In this sense II.1 is not only the first theorem of Book II but the API-level distributive kernel for the propositions that follow.

1.15 Dependency Summary

The actual formal dependency chain is

$$
\begin{gathered}
\text{Hilbert incidence + order + congruence + Group IV} \\
\Downarrow \\
\text{right angles + parallels + parallelograms + opposite-side congruence} \\
\Downarrow \\
\text{Book I scissors terms and exact triangle/parallelogram splitting} \\
\Downarrow \\
\texttt{IsRectangle} \\
\Downarrow \\
\texttt{IsRectangleContainedBy} \\
\Downarrow \\
\texttt{rectangle\_contained\_by\_unique} \\
\Downarrow \\
\texttt{rectangle\_split} \\
\Downarrow \\
\texttt{rectangle\_split\_three} \\
\Downarrow \\
\texttt{rectangle\_split\_three\_contained\_by} \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_1\_three}.
\end{gathered}
$$

The symmetric side-order branch is

$$\begin{gathered}\texttt{IsRectangleContainedBy}(PQ, RS)\\ \Downarrow\\ \text{cyclic rectangle rotation}\\ \Downarrow\\ \texttt{rectangle\_contained\_by\_swap}\\ \Downarrow\\ \texttt{rectangle\_contained\_by\_comm}.\end{gathered}$$

New geometric axiom in II.1:	no
New proposition-local axiom:	no
New rectangle abstraction:	yes, in `HilbertRectangle`
New reusable split helper:	yes
New representative-independence theorem:	yes
Exact scissors dissection used:	yes
General equicomplementability required by II.1:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Proposition-local `sorry`/`admit`:	no
Modular build:	clean

1.16 Conclusion

Proposition II.1 is the first clean instance in Book II of a product-like law being realized entirely by geometry. A segment is divided; a rectangle is cut along corresponding parallels; exact scissors equality identifies the whole with the sum of the pieces. The rectangle layer then removes the accidental choice of a concrete drawing, so Euclid's phrase "the rectangle contained by" becomes a representative-independent synthetic magnitude.

The formalization also isolates the two structural laws that make the language look algebraic. Additivity comes from actual dissection. Commutativity comes from cyclic rotation of the rectangle. A square is the special case in which the two containing segments agree. None of these steps requires a product of segment lengths or a numerical area map.

Thus the "geometric algebra" of II.1 is best understood not as hidden modern symbolism but as a calculus of geometric magnitudes whose algebraic laws emerge from constructions, congruence, and finite decomposition.

The representative-independent three-part theorem at the end of the current formalization is:

```
theorem euclid_proposition_2_1_three
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    -- Concrete dissection diagram.
    (B C D E L M K N X Y P Q : Geo.Point)
    (hRect : IsRectangle Geo B C E D)
    (hCE_PQ : Geo.Congruent C E P Q)

    -- First cut.
```

```
    (hBMC : Geo.Between B M C)
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    -- Second cut.
    (hMNC : Geo.Between M N C)
    (hLKE : Geo.Between L K E)
    (hLYC : Geo.Between L Y C)
    (hNYK : Geo.Between N Y K)
    (hMiddlePar :
      IsParallelogram Geo K L M N)
    (hFinalPar :
      IsParallelogram Geo C E K N)

    -- Arbitrary representatives of the four rectangles.
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)
    (W0 W1 W2 W3 : Geo.Point)
    (Z0 Z1 Z2 Z3 : Geo.Point)

    (hWhole :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 B C P Q)

    (hPart1 :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B M P Q)

    (hPart2 :
      IsRectangleContainedBy Geo
        W0 W1 W2 W3 M N P Q)

    (hPart3 :
      IsRectangleContainedBy Geo
        Z0 Z1 Z2 Z3 N C P Q) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       (hilbertParallelogramTerm Geo W0 W1 W2 W3 +
        hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)) := by
```

Chapter 2

Proposition II.2

2.1 Introduction

Proposition II.2 is the first place where the rectangle calculus established in II.1 is reused rather than developed further. The classical proposition is visually elementary: a square on a straight line is cut by a parallel through an arbitrary point of that line, and the two resulting rectangles fill the square. The formal reconstruction preserves exactly that geometric content, but its proof architecture is deliberately different from Euclid's.

Euclid proves II.2 directly from a square construction and a parallel through the point of division. The current Lean theorem instead imports `Proposition2_1` and applies the representative-independent binary form of II.1. One additional operation is needed: Euclid names each part rectangle with the whole side first, whereas the II.1 cut theorem returns the divided part first. The proof therefore rotates each rectangle representative by one vertex, applies II.1, and then transports the result back to Euclid's original order.

This makes II.2 a particularly clean test of the Book II API. Two algebra-like laws are already available, but neither is primitive algebra:

- additivity is realized by an exact rectangle dissection;
- exchange of the two containing segments is realized by cyclic rotation of a rectangle.

No multiplication of segment lengths and no numerical area function occurs in the theorem.

2.2 Euclid's Statement and Classical Configuration

2.2.1 The statement

In Heath's wording, Proposition II.2 states:

> If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

Here "both of the segments" means the two rectangles obtained by pairing the whole line with each of the two parts. If AB is cut at C, the geometric statement is therefore

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \;=\; \mathrm{Sq}(AB).$$

The notation is documentary shorthand only. It does not introduce a number-valued rectangle function.

2.2.2 The classical diagram

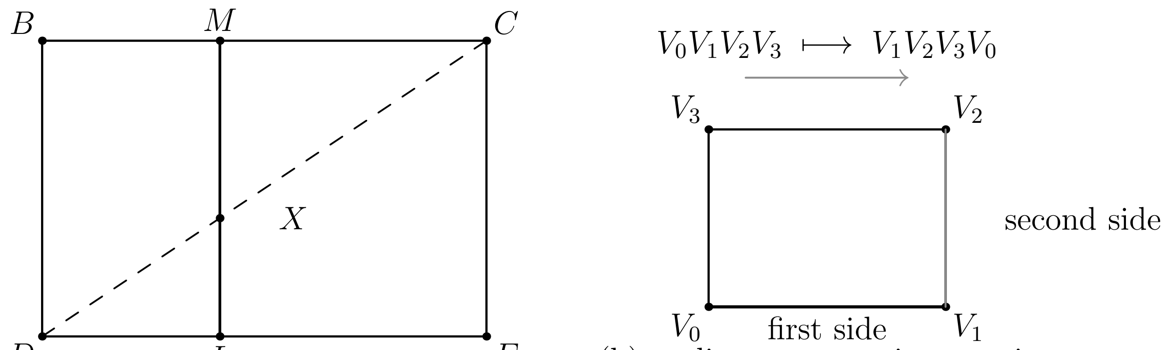
Euclid takes a straight line AB cut at C, constructs the square $ADEB$ on AB by I.46, and through C draws CF parallel to the two opposite sides AD and BE by I.31 [1]. The square is thereby partitioned into two rectangles. One is contained by the whole AB and the segment AC; the other is contained by the whole AB and the segment CB.

The current Lean configuration uses different letters:

classical role	Lean role
AB	BC
C	M
$ADEB$	$BCED$
CF	ML

Thus $B - M - C$, the outer figure $BCED$ is a square, and the cut ML divides it into the left and right rectangular pieces.

Figure 2.1 shows both the configured cut and the separate representation change used by the formal proof.



(a) square cut used by II.2
(b) cyclic representative rotation

Figure 2.1: Two mechanisms used in the formal II.2 proof. Panel (a) is the instantiated square $BCED$ with $B - M - C$ and the cut ML; the dashed diagonal DC meets the cut at the hidden witness X, so the formal data include both $D - X - C$ and $M - X - L$. Panel (b) shows the cyclic presentation change $V_0V_1V_2V_3 \mapsto V_1V_2V_3V_0$, which exchanges the two adjacent side roles while preserving exact scissors content.

2.3 Classical Proof

The classical proof has only four mathematical steps.

First, construct the square on the whole segment. Second, draw through the point of section a line parallel to the appropriate pair of sides of the square. Third, observe that this line partitions the square into two rectangles. Finally, identify those rectangles by Book II Definition 1: their adjacent sides are the whole segment together with the two parts of the divided segment.

In the standard lettering this gives

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) = \mathrm{Sq}(AB).$$

Euclid does not invoke II.1 in this proof. The local classical dependency graph is instead

$$\begin{array}{c} \text{I.46: construct the square on } AB \\ \Downarrow \\ \text{I.31: draw the parallel through } C \\ \Downarrow \\ \text{II.Def.1: identify the two rectangles} \\ \Downarrow \\ \text{the two pieces constitute the square.} \end{array}$$

This fact is important for documentation. The formal theorem proves the same geometric identity, but it is not a line-by-line formalization of Euclid's historical derivation.

2.4 The Geometric Identity

In the current Lean lettering the square side is BC and the cut point is M. The two Euclidean rectangles are therefore

$$\mathrm{Rect}(BC, BM), \qquad \mathrm{Rect}(BC, MC),$$

and the theorem asserts the exact scissors relation

$$\mathrm{Rect}(BC, BM) + \mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Sq}(BC).$$

A modern algebraic shadow is obtained by writing $x = y + z$:

$$x^2 = xy + xz.$$

This is useful as a mnemonic for the pattern, but it is not the object proved by Lean and it is not a claim that Euclid has defined multiplication on segment lengths. The formal terms remain rectangle representatives and their exact scissors relation.

2.5 Reusable Book II Infrastructure

II.2 imports

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction. Its proof consumes four pieces of the existing Book II infrastructure.

2.5.1 A square as a rectangle contained by the same side twice

From

```
hSquare : IsSquare Geo B C E D
```

the theorem

```
square_is_rectangle_contained_by_side
```

produces

```
IsRectangleContainedBy Geo B C E D B C B C
```

so the concrete square is recognized synthetically as a rectangle contained by BC, BC.

2.5.2 The binary form of II.1

The reusable distributive kernel is

```
euclid_proposition_2_1_two
```

whose geometric content is

$$\mathrm{Rect}(BC, PQ) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

when $B - M - C$ and the corresponding rectangular cut data are supplied.

For II.2 the fixed segment is specialized to $PQ = BC$. Thus the right-hand pieces naturally arrive in the order

$$\mathrm{Rect}(BM, BC), \qquad \mathrm{Rect}(MC, BC).$$

2.5.3 Exchanging the containing-side order

Euclid's statement uses the opposite order:

$$\mathrm{Rect}(BC, BM), \qquad \mathrm{Rect}(BC, MC).$$

The current proof does not treat this as "commutativity of multiplication." It calls

```
rectangle_contained_by_swap
```

on each arbitrary representative.

The theorem rotates a rectangle

$$A\,B\,C\,D \longmapsto B\,C\,D\,A,$$

returns the corresponding `IsRectangleContainedBy` fact with the two containing segments exchanged, and simultaneously returns a `HilbertScissorsEq` relating the two cyclic presentations.

The higher-level theorem `rectangle_contained_by_comm` belongs to the same rectangle API, but the public II.2 proof uses `rectangle_contained_by_swap` directly because its returned data are exactly the swapped representatives and exact scissors transports needed for the subsequent II.1 call.

2.6 Proof Architecture of the Reconstruction

The Lean proof has four stages.

2.6.1 Step 1: recognize the square

The square hypothesis is converted to the "contained by" language:

```
have hSquareContained :
    IsRectangleContainedBy Geo
      B C E D B C B C :=
  square_is_rectangle_contained_by_side
    Geo B C E D hSquare
```

The rectangle structure and the side congruence $CE \cong BC$ are then extracted from this fact.

2.6.2 Step 2: rotate the two arbitrary part representatives

The two hypotheses supplied in Euclid's order are

```
hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M

hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C
```

The current source then applies `rectangle_contained_by_swap` separately to `hPart1` and `hPart2`. Each call returns a swapped `IsRectangleContainedBy` fact and an exact scissors equality for the cyclically rotated representative. The resulting representatives are $V_1V_2V_3V_0$ and $W_1W_2W_3W_0$.

The mathematical result is that the rotated representatives now have the side orders required by the II.1 binary theorem.

2.6.3 Step 3: apply II.1 with fixed segment BC

The central call is

```
have hSplit :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B C E D)
      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
       hilbertParallelogramTerm Geo W1 W2 W3 W0) :=
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    B C
    hRect
```

```
    hCE_BC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    B C E D
    V1 V2 V3 V0
    W1 W2 W3 W0
    hSquareContained
    hPart1Swap
    hPart2Swap
```

This is the whole mathematical core of the reconstruction:

$$\mathrm{Sq}(BC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, BC) + \mathrm{Rect}(MC, BC).$$

2.6.4 Step 4: return to Euclid's ordering

The two exact rotation equalities are added:

```
have hPartsBack :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
       hilbertParallelogramTerm Geo W1 W2 W3 W0)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate1)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate2)
```

Transitivity composes `hSplit` with `hPartsBack`, and the final symmetry reverses the result into the order used by the public theorem: sum of the two Euclidean rectangles on the left, square on the right.

2.7 Lean Formalization

The current public theorem is the following instantiated diagram theorem:

```
theorem euclid_proposition_2_2
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (B C D E L M X : Geo.Point)

    (hSquare : IsSquare Geo B C E D)

    (hBMC : Geo.Between B M C)
```

```
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    (V0 V1 V2 V3 : Geo.Point)
    (W0 W1 W2 W3 : Geo.Point)

    (hPart1 :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B C B M)

    (hPart2 :
      IsRectangleContainedBy Geo
        W0 W1 W2 W3 B C M C) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo W0 W1 W2 W3)
      (hilbertParallelogramTerm Geo B C E D) := by
```

Three aspects of the signature are mathematically significant.

First, the theorem assumes an instantiated square and cut diagram. It is not an existence theorem constructing the square and the parallel cut from only a segment and a cut point. Euclid performs those constructions in the classical proof; the present public Lean theorem starts after they have been supplied.

Second, the two part rectangles are arbitrary representatives satisfying `IsRectangleContainedBy`. Their exact placement in the plane is not fixed by the statement.

Third, the conclusion is the exact relation `HilbertScissorsEq`. It is not merely a conclusion in the more general `HilbertScissorsEquicomplementable` relation. II.2 remains at the exact-dissection level because its proof is ultimately the binary exact split of II.1 plus exact representation transports.

2.8 Relationship with Proposition II.1

The contrast between the classical and formal dependency graphs is especially clear here.

Euclid's II.2 is proved directly from the square and the parallel cut. The current Lean file instead begins with

```
import CGJteamLab.Proposition2_1
```

and the public proof calls

```
euclid_proposition_2_1_two
```

once.

Thus II.2 is an explicit formal corollary of the binary II.1 API. No new dissection lemma is proved. The only additional mathematical work is conversion between the two possible orders of the containing segments.

This reuse also explains why the symmetric rectangle interface added at II.1 is not cosmetic. Without the side-order transport, the II.1 theorem would produce perfectly correct rectangles whose formal orientation does not match the language of II.2.

2.9 Relationship with Book I Infrastructure

The direct import is II.1, not a Book I proposition. Nevertheless the inherited chain underneath II.1 contains the Book I geometry on which the rectangle calculus rests.

The exact binary split is built over the scissors decomposition infrastructure developed around I.47. Rectangle structure uses Euclidean parallelogram theory and synthetic right angles; the square hypothesis uses the square infrastructure established in Book I. Consequently the formal proof of II.2 does not replay I.31 and I.46 as Euclid does. Those classical constructions have been absorbed into the established geometric API and into the hypotheses of the instantiated theorem.

2.10 Formalization Notes and Hidden Geometric Data

The visible square cut hides several obligations that are explicit parameters of the current theorem:

- $B - M - C$, fixing the orientation of the cut on the square side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the cut with the diagonal used by the inherited exact scissors split;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- two arbitrary representatives of the Euclidean part rectangles;
- two cyclic representative changes, each accompanied by an exact scissors equality.

The point X has no role in the elementary verbal statement. It belongs to the current exact-dissection interface inherited from the triangulated scissors proof. It is therefore documented as hidden formal geometry rather than presented as part of Euclid's original construction.

There is also an important negative observation. No `SameRay`, `SameSide`, or `OppositeSide` obligation appears in `Proposition2_2.lean` itself. The corresponding low-level geometry has already been discharged inside the reusable rectangle and II.1 theorems.

2.11 Geometry, Content, Representation, and Later Arithmetic

2.11.1 Geometry

The geometric layer consists of a square, betweenness on two opposite sides, the diagonal/cut witness, and two parallelogram pieces. Congruence of the square side with itself identifies the square as a rectangle contained by BC, BC.

2.11.2 Content / scissors

The conclusion and every transport used in the proof are exact `HilbertScissorsEq` relations. The proof uses symmetry, transitivity, and addition of exact scissors equalities. It does not weaken to the more general equicomplementability relation.

2.11.3 Representation

Representation is the distinctive issue in II.2. The theorem accepts arbitrary rectangles for

$$\mathrm{Rect}(BC, BM) \quad \text{and} \quad \mathrm{Rect}(BC, MC).$$

The binary II.1 theorem wants the factors in the reverse order. Cyclic rotation of each representative supplies the required conversion while preserving exact scissors content.

2.11.4 Later arithmetic / numerical area

Neither layer is used. In particular, the proof does not define a segment product, does not invoke Hilbert's later segment field, and does not assign real areas to the square or rectangles.

2.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore works in the same Euclidean project layer as II.1. No new proposition-specific geometric axiom or content axiom is declared in `Proposition2_2.lean`. Inspection of the current 162-line source shows one theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as "axiom-free." The theorem inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output was supplied with the source used for this documentation pass, so the transitive axiom list is not reproduced here.

The implementation checkpoint records that the II.2 Lean build had already been checked during theorem development. This documentation workspace does not contain the complete Lean project, so the Lean build is not re-run here; the LaTeX and Asymptote checks are reported separately.

2.13 Hartshorne's Semantic Control

Hartshorne places the end of Book I and all of Book II inside Euclid's theory of area, interpreted synthetically as equal content rather than as an unexplained numerical area function [5]. He also describes Book II as part of an "algebra of areas": rectangles can play a product-like role because Euclid has no operation that multiplies two line segments to produce another segment.

II.2 fits that interpretation exactly, provided the levels are kept separate. The square on the whole and the two rectangles are geometric figures; their equality is a content statement; the modern identity $x^2 = xy + xz$ is only an interpretive shadow.

The current Lean theorem is locally stronger than the weakest equal-content reading because its configured proof stays at exact scissors equality. That strength comes from the explicit binary dissection inherited from II.1, not from a general principle identifying every equal-content pair with an exact dissection.

2.14 Hilbert's Layering

Hilbert supplies a second semantic guardrail. His theory separates synthetic plane content from the later arithmetic of segments. Segment multiplication is introduced only in a later arithmetic construction, with a fixed unit segment and parallel geometry; it is not primitive rectangle notation.

The II.2 proof remains entirely on the earlier side of this boundary:

$$\begin{gathered}\text{square and rectangle geometry}\\ \Downarrow\\ \text{exact scissors dissection}\\ \Downarrow\\ \text{representative rotation and transport.}\end{gathered}$$

The apparent commutativity required by the formula $xy = yx$ is realized geometrically by `rectangle_contained_by_swap`, not imported from Hilbert's later segment arithmetic.

2.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{gathered}\text{Hilbert incidence + Euclidean plane layer}\\ \Downarrow\\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure}\\ \Downarrow\\ \texttt{HilbertRectangle}: \texttt{IsRectangleContainedBy}, \texttt{rectangle\_contained\_by\_swap}\\ \Downarrow\\ \text{Proposition II.1 binary API}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_1\_two}\\ \Downarrow\\ \text{two representative rotations + exact scissors addition/transitivity}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_2}.\end{gathered}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{square cut into two rectangles} \longrightarrow \text{II.2.}$$

New geometric axiom in II.2:	no
New content axiom in II.2:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.2:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Direct side-order transport used:	yes, `rectangle_contained_by_swap`
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local `sorry`/`admit`:	no

2.16 Conclusion

Proposition II.2 is mathematically a square cut into two rectangles, but formally it does more than repeat the picture. It demonstrates that the II.1 rectangle calculus is already reusable: the binary dissection theorem supplies additivity, and cyclic rectangle rotation reconciles the two possible orders of the containing sides.

This is also the first sharp divergence between Euclid's local proof graph and the formal Book II graph. Euclid proves II.2 directly from I.46 and I.31. Lean uses II.1 as a library theorem and turns II.2 into a short transport-and-reuse argument. The theorem is therefore not a literal transcription of the classical proof, but it remains entirely synthetic and geometric.

The result can be remembered by the modern shadow

$$x^2 = xy + xz, \qquad x = y + z,$$

but the formal content is more precise: arbitrary representatives of the two rectangles contained by the whole side and the two parts are exactly scissors-equal, in sum, to the concrete square on the whole side. No numerical area and no multiplication of segment lengths is required.

Appendix A

Greenberg: A Hilbert-Style Reconstruction

A.1 Purpose and scope

This appendix records the Greenberg-side source audit for the current Book II checkpoint. The present checkpoint is Proposition II.1. As in Book I, the purpose is not to force a one-to-one correspondence between Euclid's propositions and Greenberg's theorem numbering, but to identify which part of the formal architecture is controlled by Greenberg and where another source must take over.

A.2 The content boundary inherited from Book I

Greenberg distinguishes numerical area from *equal content*. He describes the latter informally through finite dissection and reassembly of polygonal figures, but explicitly declines to develop the full synthetic theory and refers the reader to Hartshorne for that purpose [4, 5].

This source boundary remains decisive at the beginning of Book II. Proposition II.1 is geometrically a rectangle dissection: one side is divided and parallels through the division points cut the large rectangle into smaller rectangles. Greenberg's Euclidean parallel and parallelogram theory is compatible with that geometric shell, but his text is not the source for the exact content calculus used by the Lean theorem.

A.3 Proposition II.1

No dedicated Greenberg proof of Euclid II.1 was located in the current source audit. What Greenberg does provide is the semantic warning needed for the formalization: Euclid's equality of plane figures should not be replaced silently by equality of real-valued areas.

The Lean reconstruction sharpens the direct-dissection case to `HilbertScissorsEq`. This should be read as project-level formal infrastructure, not as a theorem attributed to Greenberg. The division of responsibilities is therefore

Greenberg-compatible Euclidean geometry
+ Hilbert/Hartshorne content semantics
+ Lean exact scissors representation

A.4 Current checkpoint

For II.1 Greenberg is a control source for the distinction

$$\text{synthetic content} \neq \text{numerical area},$$

and for the Euclidean geometry supporting the rectangle configuration. The exact finite-dissection theorem and representative-independent rectangle API belong to the formal development itself.

Appendix B

Hartshorne: Euclid in a Hilbert Plane

B.1 Purpose and scope

Hartshorne is the principal semantic control source for the beginning of Book II. The current checkpoint is Proposition II.2. The questions are: what Euclid's rectangle language means without numerical measurement, what kind of equality of figures is involved, and in what precise sense Book II may be described as an "algebra of areas."

B.2 Book II as an algebra of areas

Hartshorne emphasizes that Euclid has no operation multiplying two line segments to obtain another line segment. Instead, a rectangle whose sides are congruent to two given segments can play a product-like geometric role. He then points to the application-of-areas propositions and to Book II as giving flexibility in manipulating and comparing such areas, producing a kind of "algebra of areas" [5].

The phrase is interpretive, not historical algebraic notation. It does not mean that Euclid has introduced variables for real lengths or a field multiplication on segments. For II.1 the modern distributive identity

$$a(b + c + d) = ab + ac + ad$$

is therefore only a mnemonic for the geometry of a rectangle cut into three subrectangles.

B.3 Proposition II.1

Hartshorne's brief Euclid summary records II.1 as the assertion that the rectangle contained by two lines is the sum of the rectangles contained by one of them and the segments of the other [5]. This formulation matches the semantic architecture of the formal theorem: one side remains fixed while the other side is divided.

The formalization deliberately separates three levels:

1. the concrete rectangle and its actual cut;

2. the phrase "rectangle contained by" two segments;

3. independence of the concrete representative chosen for that phrase.

The conclusion of the current Lean theorem is stronger than the weakest synthetic equal-content reading: it is an exact `HilbertScissorsEq`. That strengthening is justified locally because II.1 is itself a direct finite dissection. It should not be generalized into a claim that equal content and equidecomposability coincide in every later proposition.

B.4 Why numerical area is not needed

Hartshorne's area discussion is especially useful here because it keeps the synthetic and numerical layers distinct. Numerical area can be introduced later, after coordinates and numerical lengths are available, but II.1 does not require that layer. Its proof can remain entirely in the language of rectangles, congruent sides, parallels, and finite decomposition.

This is precisely the interpretation implemented by `IsRectangleContainedBy`: the two containing segments determine the adjacent side magnitudes of a representative rectangle; they are not multiplied as numbers.

B.5 Proposition II.2

Hartshorne does not isolate II.2 as a separate foundational difficulty. His general area theorem states that the propositions of Book II are valid when Euclid's equality of figures is read as equal content, and his discussion of Book II places these identities inside the same "algebra of areas" [5].

II.2 is especially transparent under that reading. A square on the whole line is divided into the two rectangles obtained by pairing the whole with the two parts. The modern mnemonic

$$x^2 = xy + xz, \qquad x = y + z,$$

describes the pattern but does not replace the geometric figures by numbers.

The current formal proof sharpens this particular instance to exact `HilbertScissorsEq`. It also exposes a representational point suppressed by the algebraic mnemonic: the ordered descriptions $\mathrm{Rect}(BC, BM)$ and $\mathrm{Rect}(BM, BC)$ are connected by cyclic rotation of a rectangle representative, not by an assumed multiplication law.

B.6 Current checkpoint

At II.2 Hartshorne continues to supply the main semantic bridge

$$\text{rectangle/square language} \longrightarrow \text{product-like geometric magnitude} \longrightarrow \text{additivity by dissection,}$$

while the Lean development makes two further distinctions explicit: representative order is handled geometrically, and the configured theorem retains exact scissors equality rather than weakening to the general equal-content relation.

Appendix C

Hilbert: The *Grundlagen* Architecture

C.1 Purpose and scope

Hilbert controls two distinct layers relevant to Book II: the synthetic theory of plane content in Chapter IV and, separately, the arithmetic of segments in Section 15. At the current checkpoint, Proposition II.2 makes the separation especially visible because its algebra-like symmetry is implemented by rectangle rotation rather than by segment multiplication.

C.2 Plane content before numerical area

Hilbert's Chapter IV distinguishes finite equidecomposability from the more general relation of equicomplementability. The theory of plane area is built on the line and plane axioms of Groups I–IV, without importing the continuity axiom at its starting point [2].

For the present reconstruction this supports the standing hierarchy

$$\text{exact finite dissection} \Longrightarrow \text{equicomplementability} \Longrightarrow \text{later numerical area semantics,}$$

without reversing either implication merely for convenience. II.1 and the current configured II.2 theorem land at the strongest first level because the underlying argument is an explicit rectangle cut together with exact representative transports.

C.3 Section 15: arithmetic of segments

Hilbert introduces an arithmetic of segments only after substantial synthetic development. Section 15 first defines segment addition from betweenness and then defines a geometric product using a fixed unit segment and a parallel construction. He subsequently proves commutativity, associativity, and the distributive law for that segment multiplication [2].

This later theory must not be read backward into the present Lean proofs. In particular, II.1 is not formalized by first defining segment multiplication and invoking a distributive identity. Additivity comes from an actual rectangle dissection, and symmetry of the two containing sides comes from cyclic rotation of a rectangle.

C.4 Proposition II.1 and the rectangle API

The formal predicate `IsRectangleContainedBy` stays below Hilbert's segment arithmetic. It says only that a concrete rectangle has adjacent sides congruent to two given segments. Representative independence is then proved through exact scissors equality.

This order of construction is significant:

$$\text{segments} + \text{congruence} + \text{right angle} \longrightarrow \text{rectangle representative} \longrightarrow \text{scissors content},$$

not

$$\text{segment field} \longrightarrow \text{numerical product} \longrightarrow \text{area formula}.$$

C.5 Proposition II.2: symmetry stays geometric

II.2 supplies a useful operational test of this separation. The binary II.1 theorem naturally produces rectangles with the divided part as the first containing segment and the whole side as the second. Euclid's II.2 statement uses the reverse order.

The Lean proof bridges the two descriptions with `rectangle_contained_by_swap`. This theorem cyclically rotates a rectangle by one vertex, simultaneously exchanging the roles of the two adjacent sides and preserving exact scissors content. Thus the formal analogue of exchanging two factors is still a theorem of rectangle geometry.

Nothing in this step invokes Hilbert's later segment product. The modern identity

$$x^2 = xy + xz$$

is therefore only a shadow of the synthetic chain

$$\text{square} \longrightarrow \text{binary rectangle dissection} \longrightarrow \text{cyclic representative transport}.$$

C.6 Current checkpoint

At II.2 Hilbert still serves two distinct documentary purposes. Chapter IV controls the content hierarchy, while the later segment arithmetic marks a boundary that the present reconstruction has not crossed. The new observation at II.2 is that even the apparent commutativity needed to match Euclid's side order is obtained below that arithmetic boundary, by geometric rotation of a rectangle representative.

Appendix D

Borsuk–Szmielew: The Axiomatic Layers

D.1 Purpose and scope

Borsuk–Szmielew are used in the project primarily as a control source for the axiomatic layering of incidence, order, congruence, polygons, parallels, and parallelograms. The current checkpoint is Proposition II.1.

D.2 Geometry and content remain separate

The source develops rigorous polygon and triangulation machinery and, in the Euclidean layer, strong parallelogram infrastructure. What it does not develop is a Euclidean synthetic theory of equal content or equicomplementability parallel to Hilbert Chapter IV or Hartshorne's treatment [3]. This boundary was already visible in the Book I area propositions and remains relevant at the opening of Book II.

For II.1 the geometric configuration itself is fully natural in this source architecture: a rectangle is a special parallelogram, its sides are controlled by Euclidean parallel geometry, and the division can be represented by polygon subdivision. The conclusion that the whole rectangle equals the sum of the subrectangles, however, belongs to the separate scissors/content layer of the formal project.

D.3 Proposition II.1

No direct Borsuk–Szmielew counterpart to Euclid II.1 was located in the current audit. The source should therefore not be cited as supplying the proof of the content identity. Its role is architectural: it confirms that figure representation and Euclidean rectangle geometry can be developed independently of any numerical area function.

This supports the formal separation

$$\text{polygon/rectangle geometry} \quad + \quad \text{scissors relation} \quad + \quad \text{representative independence.}$$

D.4 Current checkpoint

At II.1 Borsuk–Szmielew remain a geometry-and-representation control source, not the semantic source for equality of contents. This prevents the source audit from silently attributing to them a theory they do not develop.

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